

James Nithil V

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EXPERIENCE

Junior Software Developer

Aug 2025 – Present

Vaken Technology | Product: Sovablu

- Built and maintained production **REST APIs** using **Java & Spring Boot**, applied **rate limiting**, **Singleton design pattern**, and scalable system design for high-availability workflows.
- Integrated **Redis caching** at critical hotspots, reducing DB query latency, stored structured data in **PostgreSQL**.
- Managed **CI/CD pipeline** via **Jenkins** (JAR build & deploy), used **AWS CodeCommit**, **EC2**, **S3**, and **Lambda** for cloud infrastructure.
- Engineered an **Agentic AI system** using **AWS Bedrock** for LLM API calls, built **OCR pipelines** in Python to extract data from images, PDFs, and documents.
- Developed frontend features in **Vue.js** and configured **CORS & proxy** for secure client-server communication.

PROJECTS

KadalValzhi – AI Fisherman Guidance System

2026

- Built a real-time AI platform for fishermen covering **weather risk analysis**, **fish market pricing**, **fuel cost estimation**, and voyage profitability using **LangGraph multi-agent architecture**.
- Designed **RAG pipeline** with **ChromaDB** vector store using **cosine similarity search** on fishery documents for context-aware LLM responses.
- Developed **Weather**, **Financial**, **Guide**, and **LLM-as-Judge** agents; monitored agent evaluation via **LangSmith**.
- Built **LSTM time-series forecasting** model, integrated **PaddleOCR + VLM** for fish image data extraction.
- Integrated **OpenWeather API** with a **3-attempt fallback loop** across multiple weather APIs – if all fail, the system explicitly warns users rather than serving unreliable data, protecting fisherman safety.
- Applied **Singleton pattern** and **LRU caching**, logged daily fishing P&L in **PostgreSQL** for historical AI guidance.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript.

Backend : Spring Boot, FastAPI, REST APIs.

Systems : Distributed Systems, Caching, Rate Limiting, API Integration, System Design.

Databases : PostgreSQL, MySQL, Redis.

Cloud/DevOps : AWS, Docker, Jenkins, GitHub Actions.

AI/ML : LLMs, RAG, AI Agents, LangGraph, Transformers.

ACHIEVEMENTS

LeetCode Knight Badge

Global Rank #197 / 43,027

Weekly Contest 490 – Solved 4/4 Problems

LeetCode

- Earned the **LeetCode Knight** badge – one of the top competitive programming tiers on LeetCode – by achieving a **perfect 4/4 score** in Weekly Contest 490, ranking **197th out of 43,027 global participants**.
- Solved **700+ problems** total: **400+ in Java** covering Trees, Graphs, BFS/DFS, Dynamic Programming, Recursion, Two Pointers, Greedy, and Linked Lists (Single, Double, Circular); **350+ in Python3**; and **60+ in SQL & Pandas**.

EDUCATION

Knowledge Institute of Technology

Salem

B.Tech – Artificial Intelligence and Data Science | CGPA: 8.1

2021 – 2025

CERTIFICATIONS

Core Java & SQL – EBox | MySQL Database – IBM | Machine Learning – Simply Learn | Deep Learning (PyTorch) – Guvi